LECTURE 3A - FUNCTIONAL ABSTRACTION

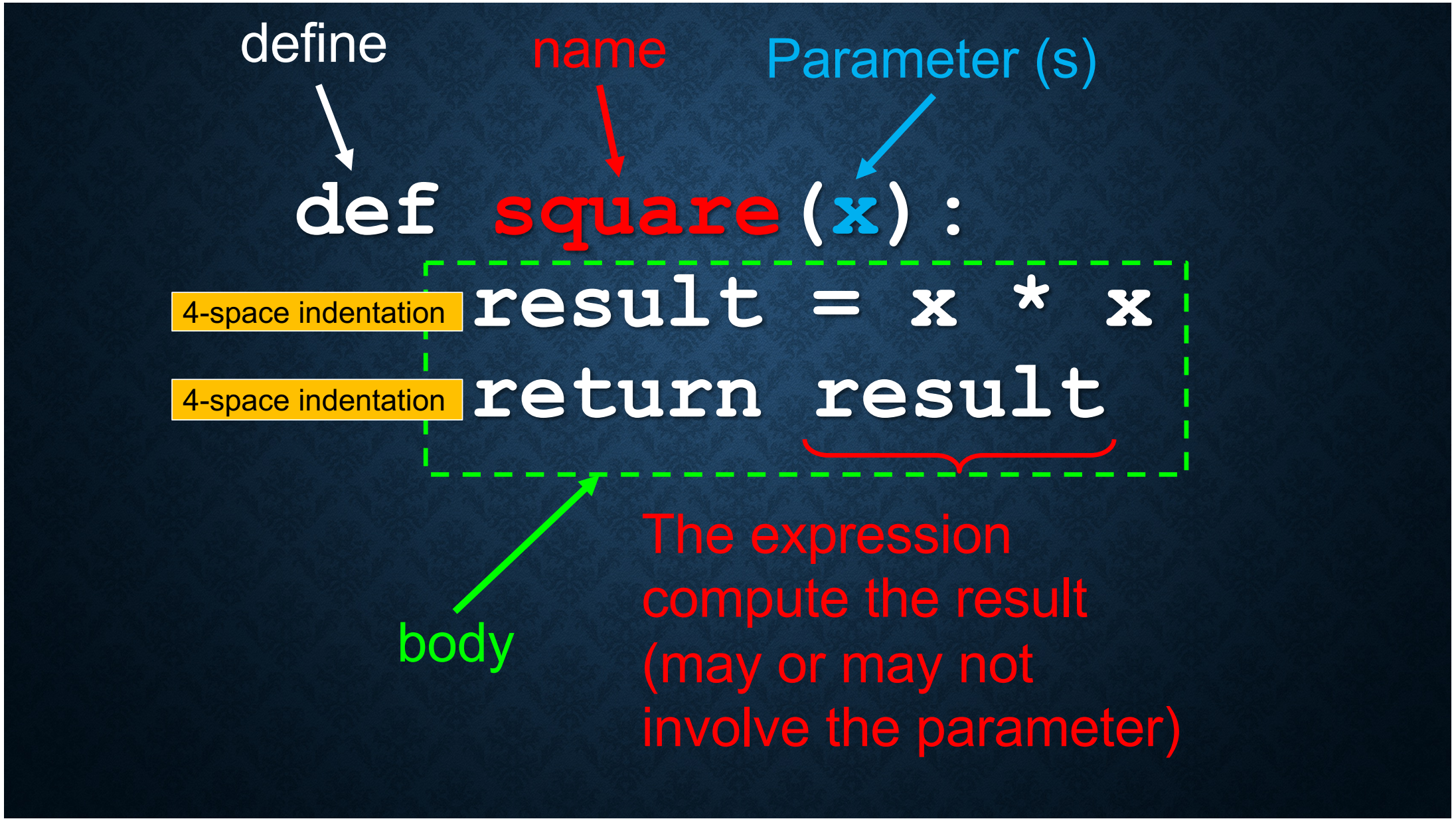
LT3A OBJECTIVES

- Understand the idea of a function in programming

General form of a function

```
def <name>(<formal parameters>):
    <body>
```

- name
  - The symbol to be associated with the function
- formal parameters
  - The names used in the body to refer to the actual arguments (e.g. values) of the function
- body
  - The expression to be evaluated to yield the result of the function application
  - Has to be indented (standard is 4 spaces)
  - Can return values as output


define
name
Parameter (s)
def square(x):
4-space indentation
result = x * x
4-space indentation
return result
body
The expression compute the result (may or may not involve the parameter)

```
square(2)
```
4

```
square(2 + 5)
```
49

```
square(square(3))
```
81

Black Box

No need to know HOW it does the job!

Just need to know WHAT it does. ☺

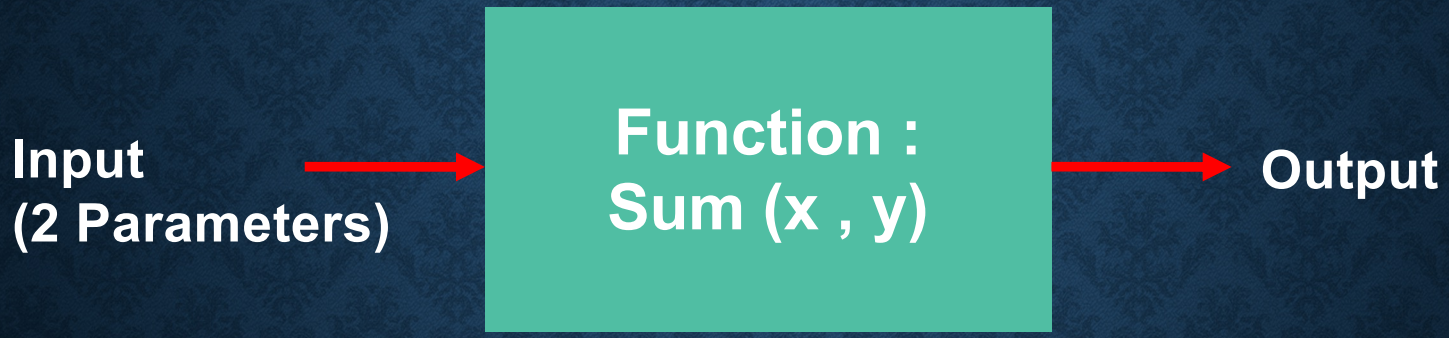
Different Implementations

```
def square(x):
    result = < expression >
    return result
```

Possible < expression > :

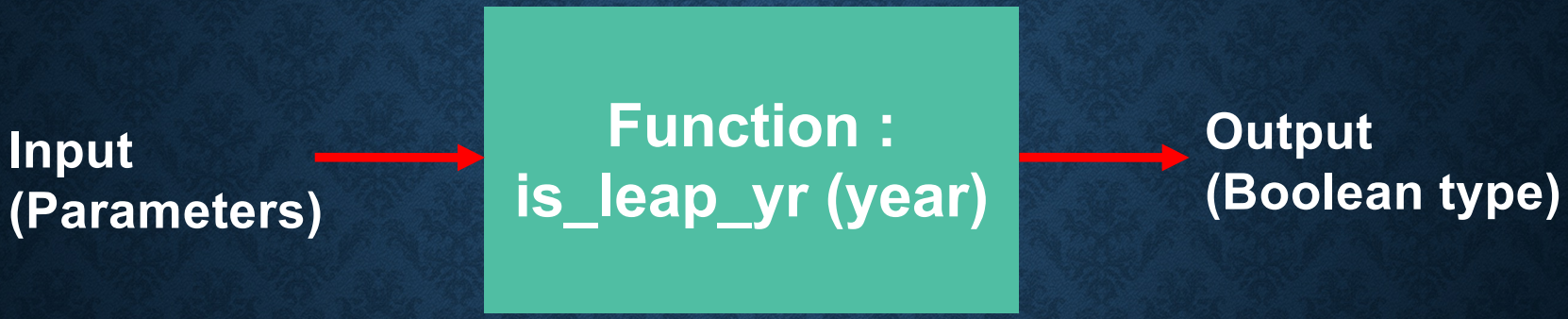
```
result = x * x

result = x ** 2

result = e ** ( 2 * ln (x))
```

Black Box



x	y	Output
1	2	3
2	3	5
3	4	7
5	6	???

Black Box



year	Output
2000	True
2100	False
2104	True
2018	???

The Output

The Output is returned with **`return`**.
The Output type could be **`None`**.

Should we use ‘print’ or ‘return’ ?

```
def square(x):
    result = x * x
    print(result)
```

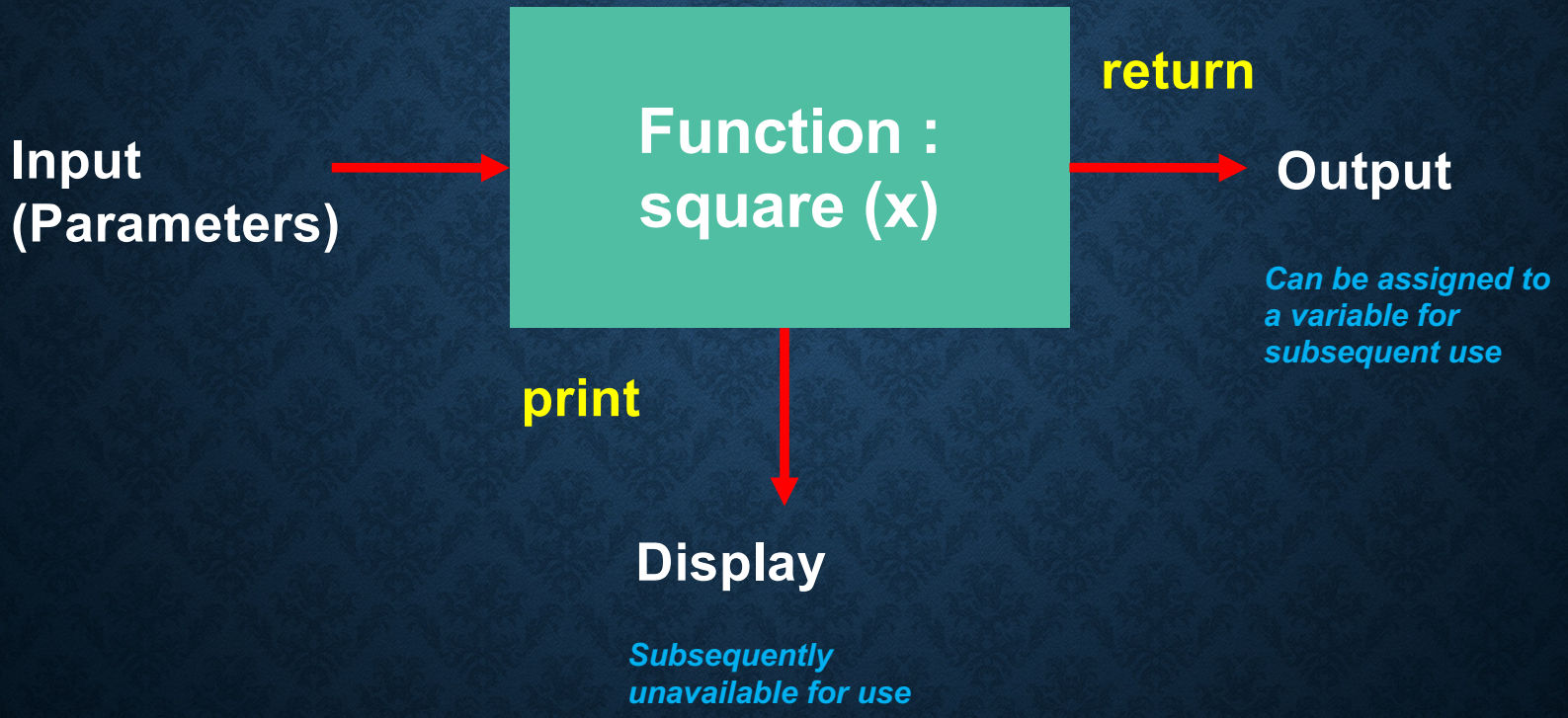
```
def square(x):
    result = x * x
    return result
```

```
Try running this :
>>> square(square(2))
```

The Output is a NoneType.

```
def square(x):
    result = x * x
    print(result)
    return None     # Python will assume this!
```

Should we use ‘print’ or ‘return’ ?



In [2]:
```
def square(x):
    result = x**2
    return result

square(2)
```

Python will forget 'result' after the function returns!

Out[2]: 4

In [3]:
```
print(result)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
<ipython-input-3-6459d04d738f> in <module>()
----> 1 print(result)

NameError: name 'result' is not defined
```

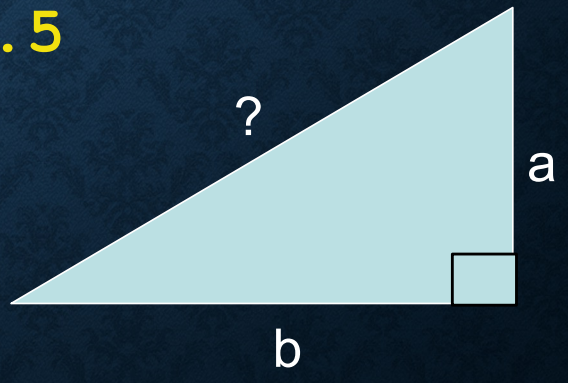
`result` is not defined

LOCAL AND GLOBAL SCOPE

Suppose you want to write a function to compute hypotenuse

This is a lousy way to code =(

```
def hypotenuse(a, b):
    return (a**2 + b**2)**0.5
```



Wishful Thinking

Top-down approach:
Pretend you have whatever you need!

$$c = \sqrt{a^2 + b^2}$$

```
def hypotenuse (a, b):
  result = sqrt (square(a) + square(b))
  return result
```

What is “sqrt”?
Pretend we know!

What is “square”?
Pretend we know!

```
def square (x):
    result = x ** 2
    return result

def sqrt (y):
    result = y ** 0.5
    return result
```

What is “square”?
Now we know!

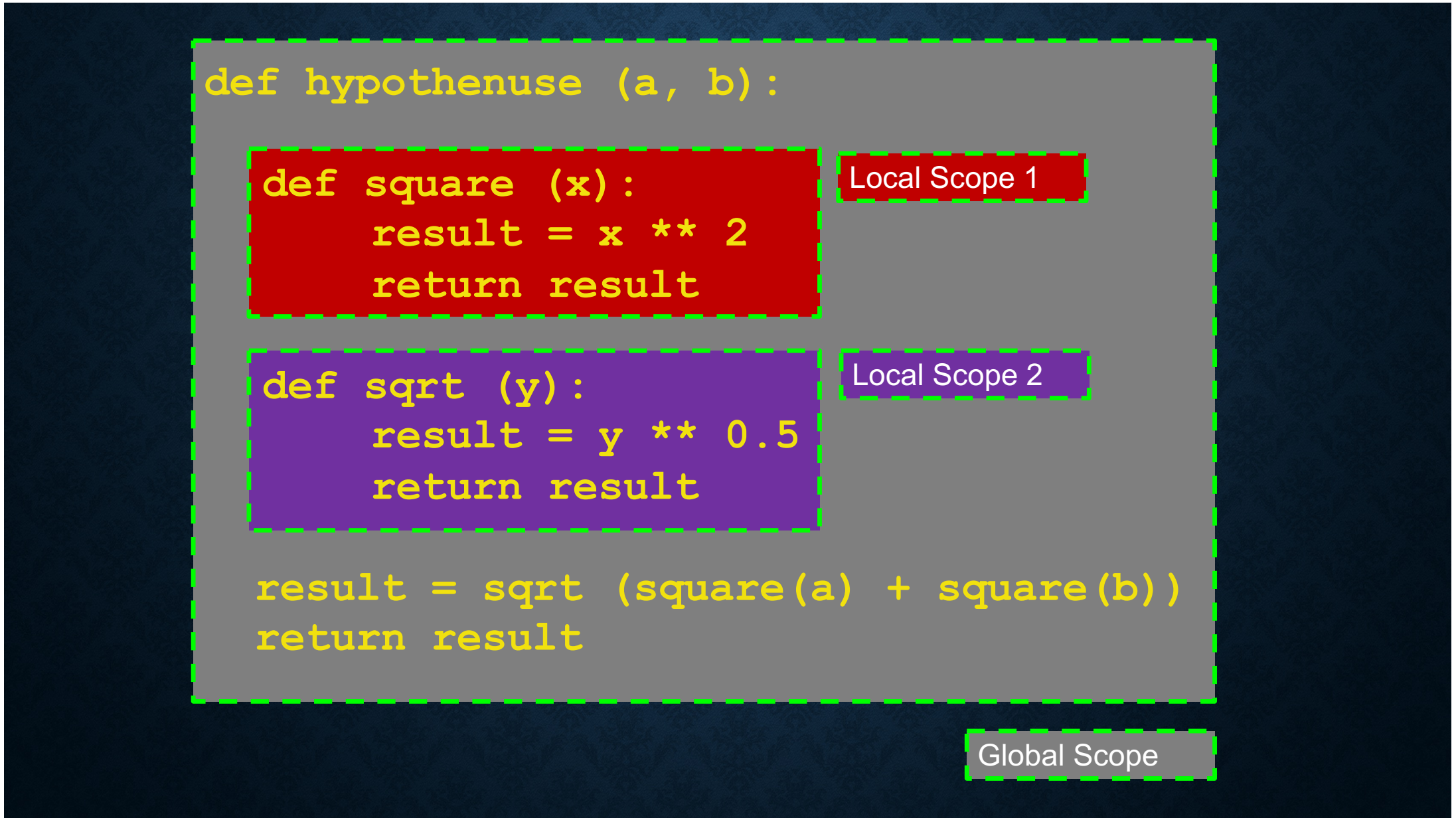
What is “sqrt”?
Now we know!

```
def square (x):
    result = x ** 2
    return result

def sqrt (y):
    result = y ** 0.5
    return result


def hypotenuse (a, b):
    result = sqrt (square(a) + square(b))
    return result
```

This is a better way to code =)

```
def hypotenuse (a, b):

    def square (x):
        result = x ** 2
        return result

    def sqrt (y):
        result = y ** 0.5
        return result

    result = sqrt (square(a) + square(b))
    return result
```

Another way of writing


def hypothenuse (a, b):
def square (x):
result = x ** 2
return result
Local Scope 1
def sqrt (y):
result = y ** 0.5
return result
Local Scope 2
result = sqrt (square(a) + square(b))
return result
Global Scope

LOCAL AND GLOBAL SCOPE

1. Code in the global scope cannot use any local variables.
2. A local scope can access global variables.
3. Code in a Function's local scope cannot use variables in any other local scope.
4. You can use the same name for different variables if they are in different scopes.

LOCAL AND GLOBAL SCOPE

1. Code in the global scope cannot use any local variables.

a and b are Global variables

```
def some_function (a, b):
    def square (x):
        result = x ** 2
        return result
    result = x + a + b
    return result
```

Local Scope 1

x is not defined

LOCAL AND GLOBAL SCOPE

2. A local scope can access global variables.

```
def another_function (a, b):
    k = 5

    def square (x):
        result = x + k
        return result

    result = square(a)
    return result
```

k is a Global variable

Local Scope 1

LOCAL AND GLOBAL SCOPE

3. Code in a Function's local scope cannot use variables in any other local scope.

```
def my_function (a, b):
    def square (x):
        result = x + y
        return result
    def sqrt (y):
        result = y ** 0.5
        return result

    result = square(a)
    return result
```

Local Scope 1

Local Scope 2

y is not defined

LOCAL AND GLOBAL SCOPE

4. You can use the same name for different variables if they are in different scopes.

```
def hypotenuse (a, b):
    def square (x):
        result = x ** 2
        return result
    def sqrt (x):
        result = x ** 0.5
        return result

    result = sqrt (square(a) + square(b))
    return result
```

Local Scope 1

Local Scope 2